

## SUMMARY

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Infrastructure and distributed-systems engineer with 12+ years designing, building, and operating foundational platforms across Kubernetes, AWS, Linux, and cloud infrastructure. I lead ambiguous, cross-functional initiatives from architecture through production, with a focus on scalable cloud abstractions, reliability, security, and developer self-service. I build infrastructure that reduces operational complexity, raises the reliability and security bar, and lets engineering teams ship faster without becoming infrastructure experts.

## SKILLS

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**Languages:** Go, Python, Bash

**Distributed Systems & Platform:** Kubernetes, Kubernetes controllers, Linux, Helm, ArgoCD, Flux, Backstage

**Cloud & Infrastructure:** AWS, GCP, Terraform, Ansible, Packer, multi-account and multi-region deployments

**Reliability & Observability:** Prometheus, Grafana, OpenTelemetry, ELK, production incident response

**Security & Delivery:** SPIRE, OPA, Vault, mTLS, Jenkins, GitHub Actions, Semaphore, Atlantis

## EXPERIENCE

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- **Confluent** Remote  
*Senior Software Engineer II, Developer Productivity* April 2025 - Present
  - **Developer platform architecture:** Co-leading technical direction for the internal developer platform, owning the developer-experience and self-service roadmap and shaping foundational capabilities that help engineering teams provision, deploy, and operate services more reliably and safely.
  - **Distributed-systems reliability:** Re-architected a production policy-enforcement service that silently dropped critical configuration-change approvals when synchronous processing exceeded an upstream timeout. Designed an asynchronous, back-pressured architecture with dashboards and alerting, eliminating the silent-failure mode and the on-call toil it caused.
  - **Infrastructure security:** Drove fleet-wide security and reliability hardening by automating service migration to zero-trust mTLS and modern least-privilege secret management, reducing operational toil while shrinking the credential attack surface.
  - **Shared infrastructure control plane:** Leading evaluation and architecture of the company's AI gateway, a shared control plane for how services access large language models. Drove cross-team technical due diligence and source-level verification that caught a supply-chain compromise in a leading candidate before adoption; an earlier infrastructure evaluation I led avoided more than \$500K in annual spend by recommending against an unready deployment.
  - **Autonomous vulnerability remediation:** Designed and shipped an AI agent that automates the end-to-end software-vulnerability remediation lifecycle; adopted as the organization's default, it remediated 300+ vulnerabilities in roughly five months. Championed its migration to a new autonomous-workflow platform and drove cross-team design reviews for the next architecture.
  - **Technical leadership:** Authored RFCs and technical one-pagers that steered cross-team platform and infrastructure decisions, mentored engineers through design and ownership transitions, and taught engineers across the organization to build reusable AI-assisted developer tooling.
- **Angi Inc** Remote  
*Senior Infrastructure Engineer* Sept 2023 - Mar 2025
  - **Cloud platform abstractions:** Built foundational components of the Angi One internal platform using Kubernetes controllers, Flux, and ArgoCD, enabling developers to provision applications and cloud services from a single declarative manifest and cutting production delivery time by 50%.
  - **Platform product experience:** Partnered with application and observability teams to automate Backstage catalog component generation and improve service onboarding, discoverability, and system visibility across the engineering organization.
  - **Developer tooling:** Led automation of documentation workflows from Kubernetes CRDs so platform documentation evolved with releases, reducing manual upkeep and support overhead.
- **Workday** Victoria, BC, Canada  
*Senior Software Development Engineer, DevOps* May 2023 - Sept 2023  
*Software Development Engineer III, DevOps* May 2021 - April 2023  
*Software Development Engineer II, DevOps* May 2019 - April 2021

- **Cloud deployment platform:** Designed an automated AWS deployment pipeline for microservices across accounts and regions, standardizing service delivery and reducing recurring operations toil by more than four hours per week.
- **Policy and self-service infrastructure:** Implemented policy-based infrastructure workflows using OPA and Atlantis, giving developers safer self-service control over cloud resources while shifting security checks earlier in the delivery lifecycle.
- **Cloud and Kubernetes infrastructure:** Built and operated production infrastructure across AWS and datacenter environments, including EC2, ECS, Route53, S3, RDS, Lambda, and Elasticsearch, while migrating applications and services onto an internal Kubernetes platform.
- **CI/CD platform:** Developed Jenkins pipelines and shared delivery libraries used by multiple applications and services, improving consistency and repeatability across build and deployment workflows.
- **Cross-team leadership:** Spearheaded an initiative across operations and development teams that helped service owners take greater responsibility for production operations; mentored engineers as they grew into larger technical responsibilities.
- **NYU CUSP** Brooklyn, NY, USA  
*Associate Research Scientist* *May 2015 - May 2019*  
*Assistant Research Scientist* *June 2014 - May 2015*
  - **Container orchestration:** Developed Dockkeeper, a scalable and secure container scheduling and monitoring system using Docker and Prometheus to provision services across physical hosts; eliminated more than \$35K per year in VM licensing costs and improved host utilization by over 55%.
  - **Distributed compute and storage infrastructure:** Architected the NYU/CUSP Urban Observatory's multi-site physical infrastructure with dense compute and more than half a petabyte of storage, provisioned as multi-user mini-HPC environments using Ansible and Packer.
  - **Production IoT platform:** Developed a secure, mission-critical IoT platform and CI/CD framework for deploying and operating more than 100 urban noise-monitoring sensors across New York City as part of the NSF-funded SONYC project.

## EDUCATION

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- **NYU Tandon School of Engineering** Brooklyn, NY, USA  
*Master of Science in Telecommunication Networks* *Aug. 2012 - May. 2014*  
*Thesis - CitySynth: Imaging with a Network of Devices*

## SELECTED PROJECTS

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- **AdguardController:** Developed a Kubernetes controller to automate DNS record management, replacing manual infrastructure changes with a declarative control loop.
- **UCSLHUB:** Developed resilient backend infrastructure using Kubernetes, JupyterHub, and Keycloak for the CUSP UCSL bootcamp, serving hundreds of students each year.
- **Homelab:** Operate a distributed HCI lab with Ansible and Terraform-managed KVM nodes, Ceph storage, Kubernetes, and network segmentation for testing distributed systems and multi-datacenter failure scenarios.

## PUBLICATIONS, TEACHING EXPERIENCE AND MORE..

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- **View it online:** <https://sharmamohit.com/>